

Qualification Pack



Technician- Distribution Transformer Repair

QP Code: PSS/Q3003

Version: 1.0

NSQF Level: 4

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PSS/Q3003: Technician- Distribution Transformer Repair

Brief Job Description

The individual at work inspects the defect/condition of distribution transformer and takes up its repair to make it functional in the work shop.

Personal Attributes

The job requires the individual to have physical strength, appropriate technical skills, ability to read, write and communicate, ability to stand for long working hours, needs to be mentally strong and demonstrate patience.

Applicable National Occupational Standards (NOS)

Compulsory NOS:

1. [PSS/N1336: Working effectively with others](#)
2. [PSS/N2001: Use basic health and safety practices for power related work](#)
3. [PSS/N3005: Testing and inspection of various faults in distribution transformer](#)
4. [PSS/N3006: Repair, overhaul and delivery of tested distribution transformer](#)

Qualification Pack (QP) Parameters

Sector	Power
Sub-Sector	Distribution
Occupation	Distribution Transformer Repair & Testing
Country	India
NSQF Level	4
Credits	NA
Aligned to NCO/ISCO/ISIC Code	NCO-2004/NIL
Minimum Educational Qualification & Experience	I.T.I (Electrician) with 2-3 Years of experience Electrician



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Minimum Level of Education for Training in School	
Pre-Requisite License or Training	Electrical maintenance training - 2 months (Suggested but not Mandatory)
Minimum Job Entry Age	20 Years
Last Reviewed On	19/07/2016
Next Review Date	31/03/2022
NSQC Approval Date	09/04/2018
Version	1.0
Reference code on NQR	2018/POW/PSSC/02186
NQR Version	1.0



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PSS/N1336: Working effectively with others

Description

This unit covers basic etiquette and competencies that a candidate is required to possess and demonstrate in their behavior and interactions with others at the workplace. These cover areas such as communication etiquette, discipline, listening, handling conflict and grievances.

Elements and Performance Criteria

working with others

To be competent, the user/individual on the job must be able to:

- PC1.** accurately receive information and instructions from the supervisor and fellow workers, getting clarification where required
- PC2.** accurately pass on information to authorized persons who require it and within agreed timescale and confirm its receipt
- PC3.** give information to others clearly, at a pace and in a manner that helps them to understand
- PC4.** display helpful behavior by assisting others in performing tasks in a positive manner, where required and possible
- PC5.** consult with and assist others to maximize effectiveness and efficiency in carrying out tasks
- PC6.** display appropriate communication etiquette while working
- PC7.** display active listening skills while interacting with others at work
- PC8.** use appropriate tone, pitch and language to convey politeness, assertiveness, care and professionalism
- PC9.** demonstrate responsible and disciplined behaviors at the workplace
- PC10.** escalate grievances and problems to appropriate authority as per procedure to resolve them and avoid conflict

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** legislation, standards, policies, and procedures followed in the company relevant to own employment and performance conditions
- KU2.** reporting structure, inter-dependent functions, lines and procedures in the work area
- KU3.** relevant people and their responsibilities within the work area
- KU4.** escalation matrix and procedures for reporting work and employment related issues
- KU5.** various categories of people that one is required to communicate and co-ordinate with in the organization
- KU6.** importance of effective communication in the workplace
- KU7.** importance of teamwork in organizational and individual success
- KU8.** various components of effective communication
- KU9.** key elements of active listening
- KU10.** value and importance of active listening and assertive communication



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- KU11.** barriers to effective communication
- KU12.** importance of tone and pitch in effective communication
- KU13.** importance of avoiding casual expletives and unpleasant terms while communicating professional circles
- KU14.** how poor communication practices can disturb people, environment and cause problems for the employee, the employer and the customer
- KU15.** importance of ethics for professional success
- KU16.** importance of discipline for professional success
- KU17.** what constitutes disciplined behavior for a working professional
- KU18.** common reasons for interpersonal conflict
- KU19.** importance of developing effective working relationships for professional success
- KU20.** expressing and addressing grievances appropriately and effectively
- KU21.** importance and ways of managing interpersonal conflict effectively

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** note the information communicated by the officer incharge
- GS2.** note down observations (if any) related to the operation/maintenance
- GS3.** read and interpret the process required for different types of manuals
- GS4.** read and interpret the flowchart of all parts of an assembly
- GS5.** read manuals and documents to understand the product-details & how they can be used
- GS6.** discuss task lists, schedules and activities with the colleague/supervisor
- GS7.** effectively communicate with the team members
- GS8.** attentively listen and comprehend the information given by the colleague/supervisor/contractor
- GS9.** communicate clearly with the colleague on the issues faced during query/fault
- GS10.** follow colleague/contractor rule-based decision making process
- GS11.** take decisions with systematic course of actions and/or response
- GS12.** planning and organization of tasks to meet deadlines
- GS13.** build customer relationships and use customer centric approach
- GS14.** seek and comprehend operation related inputs for clarification find ways of modifying difficult operating stages to make it operation friendly
- GS15.** work systematically and logically to resolve the issues and identify causation and anticipate unexpected results. Quick approach and solution towards faults repairing
- GS16.** critically evaluate operation parameters in relation to system normality develop a holistic and comprehensive profile of grid station on segregated discrete processes



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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>working with others</i>	30	70	-	-
PC1. accurately receive information and instructions from the supervisor and fellow workers, getting clarification where required	3	7	-	-
PC2. accurately pass on information to authorized persons who require it and within agreed timescale and confirm its receipt	3	7	-	-
PC3. give information to others clearly, at a pace and in a manner that helps them to understand	3	7	-	-
PC4. display helpful behavior by assisting others in performing tasks in a positive manner, where required and possible	3	7	-	-
PC5. consult with and assist others to maximize effectiveness and efficiency in carrying out tasks	3	7	-	-
PC6. display appropriate communication etiquette while working	3	7	-	-
PC7. display active listening skills while interacting with others at work	3	7	-	-
PC8. use appropriate tone, pitch and language to convey politeness, assertiveness, care and professionalism	3	7	-	-
PC9. demonstrate responsible and disciplined behaviors at the workplace	3	7	-	-
PC10. escalate grievances and problems to appropriate authority as per procedure to resolve them and avoid conflict	3	7	-	-
NOS Total	30	70	-	-



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National Occupational Standards (NOS) Parameters

NOS Code	PSS/N1336
NOS Name	Working effectively with others
Sector	Power
Sub-Sector	Generic
Occupation	Generic
NSQF Level	2
Credits	TBD
Version	1.0
Last Reviewed Date	31/03/2022
Next Review Date	31/03/2025
NSQC Clearance Date	01/03/2023



Qualification Pack

PSS/N2001: Use basic health and safety practices for power related work

Description

This unit covers health, safety and security for power related work. This includes procedures and practices that candidates need to follow to help maintain a healthy, safe and secure work environment in a power plant, power station/substation or on the field while working on power equipment. It covers responsibilities towards self, others, assets and the environment

Elements and Performance Criteria

Health and safety

To be competent, the user/individual on the job must be able to:

- PC1.** use protective clothing/equipment for specific tasks and work conditions
- PC2.** state the name and location of people responsible for health and safety in the workplace
- PC3.** state the names and location of documents that refer to health and safety in the workplace
- PC4.** identify job-site hazardous work and state possible causes of risk or accident in the workplace
- PC5.** follow electrical safe working procedures such as tag out/lock out and display ptw (permit to work),
- PC6.** follow warning signs (danger, out of service, etc.) while working with electrical systems
- PC7.** use standard safe working practices when working at heights, confined areas and trenches
- PC8.** test any electrical equipment and system using insulated testing devices before touching them
- PC9.** ensure positive isolation of electrical equipment & system as per given standards
- PC10.** recognize any abnormalities in electrical equipment or system installed alarm annunciation and/or noticing parameters from gauge/ indicator installed
- PC11.** carry out safe working practices while dealing with hazards to ensure the safety of self and others
- PC12.** state methods of accident prevention in the work environment of the job role
- PC13.** state location of general health and safety equipment in the workplace
- PC14.** inspect for faults, set up and safely use of scaffolds and elevated platforms and ladder
- PC15.** lift, carry and transport heavy objects & tools safely using correct procedures from storage to workplace and vice versa
- PC16.** inspect grid station and its equipment routinely for any signs of oil and water leakage
- PC17.** store flammable materials and machine lubricating oil safely and correctly
- PC18.** check that the emission and pollution control devices are working properly in line with environmental policy standards
- PC19.** ensure proper working condition of battery and battery charger
- PC20.** maintain electrolyte level of each cell with distilled water
- PC21.** maintain proper ventilation in battery room
- PC22.** apply good housekeeping practices at all times
- PC23.** identify common hazard signs displayed in various areas



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PC24. retrieve and/or point out documents that refer to health and safety in the workplace

PC25. inform relevant authorities about any abnormal situation/behavior of any equipment/system promptly

Fire Safety

To be competent, the user/individual on the job must be able to:

PC26. use the various appropriate fire extinguishers on different types of fires correctly

PC27. distinguish between various types of fire

PC28. demonstrate rescue techniques applied during fire hazard

PC29. demonstrate good housekeeping in order to prevent fire hazards

PC30. demonstrate the correct use of a fire extinguisher

Emergencies, rescue and first-aid procedures

To be competent, the user/individual on the job must be able to:

PC31. demonstrate how to free a person from electrocution

PC32. administer appropriate first aid to victims where required e.g. in case of bleeding, burns, choking, electric shock, poisoning etc.

PC33. demonstrate basic techniques of bandaging

PC34. respond promptly and appropriately to an accident situation or medical emergency in real or simulated environments

PC35. perform and organize loss minimization or rescue activity during an accident in real or simulated environments

PC36. administer first aid to victims in case of a heart attack or cardiac arrest due to electric shock, before the arrival of emergency services in real or simulated cases

PC37. demonstrate the artificial respiration and the cpr process

PC38. participate in emergency procedures emergency procedures: raising alarm, safe/efficient, evacuation, correct means of escape, correct assembly point, roll call, correct return to work

PC39. write accident/incident report or dictate a report to another person, and send report to person responsible

PC40. demonstrate correct method to move injured people and others during an emergency

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. legislation, standards, policies, and procedures followed in the company relevant to own employment and performance conditions

KU2. reporting structure, inter-dependent functions, lines and procedures in the work area

KU3. relevant people and their responsibilities within the work area

KU4. escalation matrix and procedures for reporting work and employment related issues

KU5. meaning of hazards and risks

KU6. health and safety hazards commonly present in the work environment and related precautions

KU7. possible causes of risk, hazard or accident in the workplace and why risk and/or accidents are possible



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- KU8.** possible causes of risk and accident
- KU9.** methods of accident prevention
- KU10.** safe working practices when working with tools and machines
- KU11.** safe working practices while working at various hazardous sites
- KU12.** where to find all the general health and safety equipment in the workplace
- KU13.** various dangers associated with the use of electrical equipment
- KU14.** positive isolation of electrical equipment and system
- KU15.** safe handling and disposal of hazardous power plant wastes
- KU16.** use of emission and pollution control devices and measures taken to control pollution
- KU17.** various safety procedures and equipment used to work at heights, trenches and confined places
- KU18.** safe working practices specific to working with electrical equipment & system e.g. lock out/tag out, ptw, etc.
- KU19.** preventative and remedial actions to be taken in the case of exposure to toxic materials
- KU20.** importance of using protective clothing/equipment and other insulated work gear while handling electrical system and equipment
- KU21.** precautionary activities taken to prevent fire accident
- KU22.** various causes of fire
- KU23.** techniques of using the different fire extinguishers
- KU24.** different methods of extinguishing fire
- KU25.** different materials used for extinguishing fire
- KU26.** emergency rescue techniques applied during a fire hazard
- KU27.** various types of safety signs and what they mean
- KU28.** appropriate basic first aid treatment relevant to the condition e.g. shock, electrical shock, bleeding, breaks to bones, minor burns, resuscitation, poisoning, eye injuries

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** note the information communicated by the officer incharge.
- GS2.** note down observations (if any) related to the operation/maintenance.
- GS3.** read and interpret the process required for different types of manuals for maintenance.
- GS4.** read and interpret the flowchart of all parts of an assembly.
- GS5.** read manuals and documents to understand the product-details & how they can be used.
- GS6.** discuss task lists, schedules and activities with the colleague/supervisor.
- GS7.** effectively communicate with the team members.
- GS8.** attentively listen and comprehend the information given by the colleague/supervisor/contractor.
- GS9.** communicate clearly with the colleague on the issues faced during query/fault.
- GS10.** follow colleague/contractor rule-based decision making process.
- GS11.** take decisions with systematic course of actions and/or response.



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- GS12.** planning and organization of tasks to meet deadlines.
- GS13.** build customer relationships and use customer centric approach.
- GS14.** seek and comprehend operation related inputs for clarification
- GS15.** find ways of modifying difficult operating stages to make it operation friendly
- GS16.** work systematically and logically to resolve the issues and identify causation and anticipate unexpected results.
- GS17.** quick approach and solution towards faults repairing.
- GS18.** critically evaluate operation parameters in relation to system normality
- GS19.** develop a holistic and comprehensive profile of grid station on segregated discrete process.

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Health and safety</i>	11	46	-	-
PC1. use protective clothing/equipment for specific tasks and work conditions	-	3	-	-
PC2. state the name and location of people responsible for health and safety in the workplace	-	2	-	-
PC3. state the names and location of documents that refer to health and safety in the workplace	-	2	-	-
PC4. identify job-site hazardous work and state possible causes of risk or accident in the workplace	1	1	-	-
PC5. follow electrical safe working procedures such as tag out/lock out and display ptw (permit to work),	1	2	-	-
PC6. follow warning signs (danger, out of service, etc.) while working with electrical systems	1	1	-	-
PC7. use standard safe working practices when working at heights, confined areas and trenches	1	2	-	-
PC8. test any electrical equipment and system using insulated testing devices before touching them	1	2	-	-
PC9. ensure positive isolation of electrical equipment & system as per given standards	1	2	-	-
PC10. recognize any abnormalities in electrical equipment or system installed alarm annunciation and/or noticing parameters from gauge/ indicator installed	1	2	-	-
PC11. carry out safe working practices while dealing with hazards to ensure the safety of self and others	1	1	-	-
PC12. state methods of accident prevention in the work environment of the job role	-	2	-	-
PC13. state location of general health and safety equipment in the workplace	-	2	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC14. inspect for faults, set up and safely use of scaffolds and elevated platforms and ladder	-	2	-	-
PC15. lift, carry and transport heavy objects & tools safely using correct procedures from storage to workplace and vice versa	-	2	-	-
PC16. inspect grid station and its equipment routinely for any signs of oil and water leakage	-	2	-	-
PC17. store flammable materials and machine lubricating oil safely and correctly	-	2	-	-
PC18. check that the emission and pollution control devices are working properly in line with environmental policy standards	1	1	-	-
PC19. ensure proper working condition of battery and battery charger	1	1	-	-
PC20. maintain electrolyte level of each cell with distilled water	-	2	-	-
PC21. maintain proper ventilation in battery room	-	1	-	-
PC22. apply good housekeeping practices at all times	1	2	-	-
PC23. identify common hazard signs displayed in various areas	-	2	-	-
PC24. retrieve and/or point out documents that refer to health and safety in the workplace	-	2	-	-
PC25. inform relevant authorities about any abnormal situation/behavior of any equipment/system promptly	-	3	-	-
Fire Safety	4	9	-	-
PC26. use the various appropriate fire extinguishers on different types of fires correctly	1	1	-	-
PC27. distinguish between various types of fire	1	2	-	-
PC28. demonstrate rescue techniques applied during fire hazard	1	2	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC29. demonstrate good housekeeping in order to prevent fire hazards	-	2	-	-
PC30. demonstrate the correct use of a fire extinguisher	1	2	-	-
<i>Emergencies, rescue and first-aid procedures</i>	9	21	-	-
PC31. demonstrate how to free a person from electrocution	1	2	-	-
PC32. administer appropriate first aid to victims where required e.g. in case of bleeding, burns, choking, electric shock, poisoning etc.	-	3	-	-
PC33. demonstrate basic techniques of bandaging	1	2	-	-
PC34. respond promptly and appropriately to an accident situation or medical emergency in real or simulated environments	1	2	-	-
PC35. perform and organize loss minimization or rescue activity during an accident in real or simulated environments	1	2	-	-
PC36. administer first aid to victims in case of a heart attack or cardiac arrest due to electric shock, before the arrival of emergency services in real or simulated cases	1	2	-	-
PC37. demonstrate the artificial respiration and the cpr process	1	2	-	-
PC38. participate in emergency procedures emergency procedures: raising alarm, safe/efficient, evacuation, correct means of escape, correct assembly point, roll call, correct return to work	1	2	-	-
PC39. write accident/incident report or dictate a report to another person, and send report to person responsible	1	2	-	-
PC40. demonstrate correct method to move injured people and others during an emergency	1	2	-	-
NOS Total	24	76	-	-



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National Occupational Standards (NOS) Parameters

NOS Code	PSS/N2001
NOS Name	Use basic health and safety practices for power related work
Sector	Power
Sub-Sector	Generic
Occupation	Generic
NSQF Level	2
Credits	TBD
Version	1.0
Last Reviewed Date	31/03/2022
Next Review Date	14/04/2024
NSQC Clearance Date	31/03/2022



Qualification Pack

PSS/N3005: Testing and inspection of faults in Distribution transformer

Description

This unit covers the ability and knowledge required by Technician: DistributionTransformer Repair to test all types of distribution transformer and other associated components. It also includes checking and inspection of smell, visual, sensory, noise, humming and vibrations in the DT. This also includes carrying out necessary testing in a safe, efficient and effective manner.

Scope

This unit/task covers the following:

- testing and inspection, identification of faults and root cause of DT failure

Elements and Performance Criteria

Testing and inspection, identification of faults and root cause of DT failure

To be competent, the user/individual on the job must be able to:

- PC1.** maintain a record card which contains the basic information of a DT like serial number, diagram, rating plate and other related aspects
- PC2.** maintain defect/repair record card which shows diagnostic records to assess the dt performance history
- PC3.** prepare check list of parameters to be kept in to consideration while doing testing and inspection of distribution transformer
- PC4.** checking general appearance and leakage of oil to identify visual faults
- PC5.** identify the nature of fault and damage of part/ component
- PC6.** disconnect the winding connections from terminal bushing and earth connection between core and tank before lifting
- PC7.** inspect physical condition visually for rust on body and on radiators
- PC8.** verify correct connections of HT/LT side
- PC9.** inspect bolt/lugs and solder of electrical connections
- PC10.** inspect all required grounding and shorting connections, perform insulation-resistance test
- PC11.** check the oil level in oil cap under silica gel breather
- PC12.** check bushing collar, gaskets and gasket joints for any leakage of oil
- PC13.** check breathing holes in silica gel breather
- PC14.** inspect color of silica gel in breather
- PC15.** check condition of OLTC
- PC16.** check leakage from gasket, gasket joints and flanges
- PC17.** inspect porcelain insulator bushing for any damage, flash and hair crack
- PC18.** identify faults arising due to: primary winding burnt (one phase, two phase or complete), braze /solder of LT winding joints melted, over heat, open circuit in internal wiring etc.
- PC19.** detect/ trouble shooting of excess humming noise due to loose fitting of silicon mixed steel alloys laminated core joints



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Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** process standards and procedures followed in the organization
- KU2.** know the persons and the responsibilities within the work area
- KU3.** work area and number of distribution transformer under the area
- KU4.** proper care of inventory management, quality management
- KU5.** keep proper documentation, records and related procedures applicable
- KU6.** know the employee related rules and regulations
- KU7.** principles of electricity
- KU8.** rules and procedures of safety
- KU9.** how and where to keep record, accessories, gadgets, equipments, PPEs, tools & tackles systematic to maintain good house keeping
- KU10.** job responsibilities/duties and standard inspection procedures
- KU11.** related power system aspects like ratings and various types of DT
- KU12.** working of a dt, its component, accessories and their functioning. difference between dry type and oil immersed transformer and their usage at site
- KU13.** various material / parts / accessories required for maintenance like insulating oil, type of core, winding material, bushings, cable and conductor, cabling box, cooling radiators, conservators, oil gauges, valves, explosion vents or pressure release devices, sealing gaskets, temperature indicators, poles, insulators and fuses etc.
- KU14.** use of tools and kits required for testing, repair and maintenance ,OLTC Continuity & Resistance Measurement Test, Dissolve Gas Analyzer Kit, temperature monitor device, Partial Discharge Test set, Flash point Test set of Oil, discharge rod, chain pulley, tripod, crane, hoist, force pulley with sling, tommy bar, crimping machine, drilling machine, meggar, tong tester etc
- KU15.** procedure and technical requirements for testing, repair and maintenance of the distribution transformer
- KU16.** reasons of major faults occurred in transformer and their cause
- KU17.** transformer winding, placing various types of insulations, fitting of core joints and complete assembly
- KU18.** testing of the performance and condition of distribution transformer
- KU19.** safety at work at all times, complying with health, safety and other relevant regulations and guidelines
- KU20.** importance of reporting problem to junior engineer (Supervisor)
- KU21.** reporting of any hazards identified and any actions taken
- KU22.** identification of any potential hazards and take appropriate action to minimize the risk
- KU23.** testing and inspection within agreed time scales using approved testing equipment, materials, components, methods and procedures

Generic Skills (GS)

User/individual on the job needs to know how to:



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- GS1.** communicate effectively in writing
- GS2.** be able to write the information communicated by the in-charge of work
- GS3.** write properly about the technical problems and other conditions of site
- GS4.** note down of testing repair observations, critical points
- GS5.** be able to write about the condition of equipment
- GS6.** prepare and fill up all technical forms and data as per guidelines and format
- GS7.** read and understand written sentences and paragraphs
- GS8.** read metric system for all measurements
- GS9.** interpret the process required for performing of work
- GS10.** read, interpret and understand the rules and methods
- GS11.** read equipment manuals and understand the equipment operation and process requirement
- GS12.** effectively communicate verbally
- GS13.** be able to communicate effectively with voice modulation, tone of voice and eye contact
- GS14.** use good body language for good oral communication
- GS15.** discuss task lists, schedules and activities with the junior engineer
- GS16.** effectively communicate with the team/group members
- GS17.** listen the information given by the junior engineer
- GS18.** able to communicate clearly with the team and other staff
- GS19.** make work related judgments appropriately
- GS20.** identifying complex problems and review related information to develop and evaluate
- GS21.** follow organization rule based decision making process
- GS22.** take decision with systematic course of actions and/or response
- GS23.** planning and organization of tasks to meet deadlines
- GS24.** identify problems and review related information to develop, evaluate options and implement solutions
- GS25.** prioritize and plan for solving problem
- GS26.** take help from the junior engineer to solve the problems
- GS27.** implement corrective action with individuals and organizations for problem solving
- GS28.** analyze problems and changes in conditions, operations, and the environment to solve problems
- GS29.** analyze the problem seen in the equipment
- GS30.** collect the information and technical data and define process for doing testing and maintenance
- GS31.** critically evaluate operation parameters in relation to distribution transformer features intended
- GS32.** develop holistic and comprehensive profile of distribution transformer repair

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Testing and inspection, identification of faults and root cause of DT failure</i>	22	78	-	-
PC1. maintain a record card which contains the basic information of a DT like serial number, diagram, rating plate and other related aspects	1	4	-	-
PC2. maintain defect/repair record card which shows diagnostic records to assess the dt performance history	1	4	-	-
PC3. prepare check list of parameters to be kept in to consideration while doing testing and inspection of distribution transformer	2	4	-	-
PC4. checking general appearance and leakage of oil to identify visual faults	1	5	-	-
PC5. identify the nature of fault and damage of part/ component	2	6	-	-
PC6. disconnect the winding connections from terminal bushing and earth connection between core and tank before lifting	1	4	-	-
PC7. inspect physical condition visually for rust on body and on radiators	-	4	-	-
PC8. verify correct connections of HT/LT side	1	4	-	-
PC9. inspect bolt/lugs and solder of electrical connections	1	4	-	-
PC10. inspect all required grounding and shorting connections, perform insulation-resistance test	2	3	-	-
PC11. check the oil level in oil cap under silica gel breather	2	4	-	-
PC12. check bushing collar, gaskets and gasket joints for any leakage of oil	1	4	-	-
PC13. check breathing holes in silica gel breather	1	4	-	-
PC14. inspect color of silica gel in breather	-	4	-	-



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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC15. check condition of OLTC	2	4	-	-
PC16. check leakage from gasket, gasket joints and flanges	-	4	-	-
PC17. inspect porcelain insulator bushing for any damage, flash and hair crack	1	4	-	-
PC18. identify faults arising due to: primary winding burnt (one phase, two phase or complete), braze /solder of LT winding joints melted, over heat, open circuit in internal wiring etc.	2	4	-	-
PC19. detect/ trouble shooting of excess humming noise due to loose fitting of silicon mixed steel alloys laminated core joints	1	4	-	-
NOS Total	22	78	-	-



Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	PSS/N3005
NOS Name	Testing and inspection of faults in Distribution transformer
Sector	Power
Sub-Sector	Distribution
Occupation	Distribution Transformer Repair & Testing
NSQF Level	4
Credits	TBD
Version	1.0
Last Reviewed Date	19/07/2016
Next Review Date	31/03/2022
NSQC Clearance Date	09/04/2018



Qualification Pack

PSS/N3006: Repair, Overhaul and Delivery of tested distribution transformer

Description

This unit covers the ability and knowledge required by Technician - distribution transformer repair to repair all types of distribution transformer and other associated components. The Technician - distribution transformer repair will be expected to perform and act independently for following ratings: i) the standard voltage rating of 11/0.433 KV ii) Capacity Range of DT is 10 KVA, 16KVA, 25 KVA, 63KVA, 100KVA, 160KVA, 200KVA, 250KVA, 315KVA, 400 KVA, 500KVA, 630KVA 750KVA, 1000KVA.

Scope

This unit/task covers the following:

- prepare and carrying out repair of defective distribution transformer
- testing and checking of over hauled distribution transformer before delivery

Elements and Performance Criteria

Prepare and carrying out repair of defective distribution transformer

To be competent, the user/individual on the job must be able to:

- PC1.** demonstrate repair and maintenance compliance stated in the standard procedure manual
- PC2.** refer maintenance manual and circuit diagram
- PC3.** ensure all required tools and kits are in good condition
- PC4.** check that all testing kits are calibrated
- PC5.** record all the abnormalities and defects during repair
- PC6.** prepare work area as per standard repair procedure
- PC7.** ensure that adequate spare parts should be kept on hand to replace the faulty parts
- PC8.** take oil samples from tank bottom, tank top and radiator for checking of break-down voltage (bdv) test
- PC9.** remove core and windings from the tank for visual inspection
- PC10.** ensure core and winding in proper cover, dry and safe place after removal from tank
- PC11.** check status of core, primary winding, secondary winding, primary terminal connections, secondary terminal connections, insulation (fish paper, empire tape/cloth, wooden spacers, tags etc)
- PC12.** identify nature of fault and carry out repair and replacement
- PC13.** place complete core and winding block for heat treatment in vacuum chamber
- PC14.** maintain voltage within prescribed limits by the use of an Off-Circuit Tap Selector (OCTS)
- PC15.** test for variation appearing in the primary side supply voltage and the secondary side supply voltage
- PC16.** check insulation resistance by megger
- PC17.** check all loose bolts / screws / clamps, tighten the core joints, solder ht and lt terminal connections



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- PC18.** check and ensure that no sludge has been deposited on winding to block the oil ducts and opening passage
- PC19.** check indoor and outdoor bushings for oil leakage and cracks or any other defects, replace the defective bushing
- PC20.** check cooling radiators for any oil leakages along all the welded joints, gasket joints and plugs. rectify the same from the radiators
- PC21.** check and ensure clamping of the conservator
- PC22.** check and clean all the oil gauges and replace the defective oil gauges
- PC23.** check the dehydrating breather and replace if saturated and color has changed
- PC24.** check that no foreign items have been left in the tank
- PC25.** repair oil leakage and sweating. top-up oil as per instruction stated in the manual
- PC26.** check pressure release device and explosion vent
- PC27.** check sealing gaskets for cracks, tight nut and bolts and replace damage gaskets
- PC28.** check oil level in conservator tank gauge and thermometer
- PC29.** check OITC switch for arcing welding and wearing and replace repair defective parts
- PC30.** check and clean the radiator with compressed air or water
- PC31.** check arcing horns for dent, welds or any defect and replace the same if found defective
- PC32.** check for any rust and damage of paint for external tank
- PC33.** check oil temperature indicator (OTI) and winding temperature indicator (WTI)
- PC34.** check air-release plugs of main tank, radiator, conservator, bushings, etc. are free of air pocket / bubbles
- PC35.** energize distribution transformer at NO-LOAD only and checked for any abnormalities for the next 4 to 8 hours
- PC36.** take advice from the manufacturer or suppliers if any major abnormalities or defects found during repair and maintenance

Testing and checking of over hauled Distribution transformer before delivery

To be competent, the user/individual on the job must be able to:

- PC37.** ensure complete transformer with its components are fitted and packed in its original shape
- PC38.** confirm all the test are done before delivery. all the test relevant to the performance of DT and ensure basic parameters like physical: leakage, low oil, silica in breather, HV & LV bushing. electrical: IR Value (HT to e, IT to e, HT to IT, Oil BDV)
- PC39.** ensure vent pipe is sealed with aluminum foil (diaphragm), temperature gauge is fitted and all HV terminals are fitted with horn and double screws and washers
- PC40.** check list before delivery: oil level, no leakage of oil, tap position, silica gel in breather, radiator valve, thermometer packet, earth connection
- PC41.** ensure that the inspected and tested component meets the specified operating conditions before issue of ok certificate
- PC42.** anticipate problems well in advance in order to rectify timely

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:



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- KU1.** process standards and procedures followed in the organization
- KU2.** know the persons and the responsibilities within the work area
- KU3.** work area and number of distribution transformer under the area
- KU4.** proper care of inventory management, quality management
- KU5.** keep proper documentation, records and related procedures applicable
- KU6.** know the employee related rules and regulations
- KU7.** principles of electricity
- KU8.** rules and procedures of safety
- KU9.** how and where to keep record, accessories, gadgets, equipments, PPEs, tools & tackles systematic to maintain good house keeping
- KU10.** related power system aspects like ratings and various types of DT transformer, its component, accessories and their functioning. difference between dry type and oil immersed transformer and their usage at site
- KU11.** various material / parts / accessories required for maintenance like insulating oil, type of core, winding material, bushings, cable and conductor, cabling box, cooling radiators, conservators, oil gauges, valves, explosion vents or pressure release devices, sealing gaskets, temperature indicators, poles, insulators and fuses etc.
- KU12.** usage of tools and kits required for testing, repair and maintenance : OLTC Continuity & Resistance Measurement Test, Dissolve Gas Analyzer kit, temperature monitor device, Partial Discharge Test Set, Flash Point test set of Oil, discharge rod, chain pulley, tripod, crane, hoist, force pulley with sling, tommy bar, crimping machine, drilling machine, meggar, tong tester,
- KU13.** procedure and technical requirements for testing, repair and maintenance of the distribution transformer
- KU14.** transformer winding, placing various types of insulations, fitting of core joints and complete assembly
- KU15.** heat treatment methods, temperature control in oven and operation of vacuum chamber
- KU16.** operations of transformer oil filter machine
- KU17.** keeping records of calibration schedule of equipment kits used for inspection, repair and maintenance
- KU18.** test the performance and condition of distribution transformer,
- KU19.** work safely at all times, complying with health, safety and other relevant regulations and guidelines.
- KU20.** importance of reporting problem to junior engineer (Supervisor)
- KU21.** report any hazards identified and any actions taken
- KU22.** identify any potential hazards and take appropriate action to minimize the risk

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** communicate effectively in writing
- GS2.** able to write the information communicated by the in-charge of work
- GS3.** write properly about the technical problems and other conditions of site



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- GS4.** note down of testing repair observations, critical points
- GS5.** able to write about the condition of equipment
- GS6.** prepare and fill up all technical forms and data as per guidelines and format
- GS7.** reading, understanding of written sentences and paragraphs
- GS8.** able to read metric system for all measurements
- GS9.** interpret the process required for performing of work
- GS10.** read, interpret and understand the rules and methods
- GS11.** read equipment manuals and understand the equipment operation and process requirement
- GS12.** effective verbal communication
- GS13.** able to communicate effectively with voice modulation, tone of voice and eye contact
- GS14.** use good body language for good oral communication
- GS15.** discuss task lists, schedules and activities with the junior engineer
- GS16.** effectively communicate with the team/group members
- GS17.** listen the information given by the junior engineer
- GS18.** able to communicate clearly with the team and other staff
- GS19.** make work related judgments appropriately
- GS20.** identifying complex problems and review related information to develop and evaluate
- GS21.** follow organization rule based decision making process
- GS22.** take decision with systematic course of actions and/or response
- GS23.** planning and organization of tasks to meet deadlines
- GS24.** identify problems and review related information to develop and evaluate options and implement solutions
- GS25.** prioritize and plan for solving problem
- GS26.** take help from the junior engineer to solve the problems
- GS27.** monitor problem solving to take corrective action with individuals and organizations
- GS28.** analyze problems and changes in conditions, operations, and the environment to solve problems
- GS29.** analyze the problem seen in the equipment
- GS30.** collect the information and technical data and define process for doing testing and maintenance
- GS31.** critically evaluate operation parameters in relation to distribution transformer features intended
- GS32.** develop holistic and comprehensive profile of distribution transformer repair based on segregated discrete process stages

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Prepare and carrying out repair of defective distribution transformer</i>	19	62	-	-
PC1. demonstrate repair and maintenance compliance stated in the standard procedure manual	1	2	-	-
PC2. refer maintenance manual and circuit diagram	1	1	-	-
PC3. ensure all required tools and kits are in good condition	-	2	-	-
PC4. check that all testing kits are calibrated	1	1	-	-
PC5. record all the abnormalities and defects during repair	1	1	-	-
PC6. prepare work area as per standard repair procedure	1	1	-	-
PC7. ensure that adequate spare parts should be kept on hand to replace the faulty parts	-	2	-	-
PC8. take oil samples from tank bottom, tank top and radiator for checking of break-down voltage (bdv) test	1	3	-	-
PC9. remove core and windings from the tank for visual inspection	-	2	-	-
PC10. ensure core and winding in proper cover, dry and safe place after removal from tank	1	2	-	-
PC11. check status of core, primary winding, secondary winding, primary terminal connections, secondary terminal connections, insulation (fish paper, empire tape/cloth, wooden spacers, tags etc)	-	2	-	-
PC12. identify nature of fault and carry out repair and replacement	1	2	-	-
PC13. place complete core and winding block for heat treatment in vacuum chamber	1	3	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC14. maintain voltage within prescribed limits by the use of an Off-Circuit Tap Selector (OCTS)	1	2	-	-
PC15. test for variation appearing in the primary side supply voltage and the secondary side supply voltage	1	1	-	-
PC16. check insulation resistance by megger	1	1	-	-
PC17. check all loose bolts / screws / clamps, tighten the core joints, solder ht and lt terminal connections	-	2	-	-
PC18. check and ensure that no sludge has been deposited on winding to block the oil ducts and opening passage	-	1	-	-
PC19. check indoor and outdoor bushings for oil leakage and cracks or any other defects, replace the defective bushing	-	2	-	-
PC20. check cooling radiators for any oil leakages along all the welded joints, gasket joints and plugs. rectify the same from the radiators	-	2	-	-
PC21. check and ensure clamping of the conservator	-	1	-	-
PC22. check and clean all the oil gauges and replace the defective oil gauges	-	2	-	-
PC23. check the dehydrating breather and replace if saturated and color has changed	1	2	-	-
PC24. check that no foreign items have been left in the tank	-	1	-	-
PC25. repair oil leakage and sweating. top-up oil as per instruction stated in the manual	1	1	-	-
PC26. check pressure release device and explosion vent	-	2	-	-
PC27. check sealing gaskets for cracks, tight nut and bolts and replace damage gaskets	-	2	-	-
PC28. check oil level in conservator tank gauge and thermometer	-	2	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC29. check OITC switch for arcing welding and wearing and replace repair defective parts	-	2	-	-
PC30. check and clean the radiator with compressed air or water	-	1	-	-
PC31. check arcing horns for dent, welds or any defect and replace the same if found defective	-	2	-	-
PC32. check for any rust and damage of paint for external tank	-	1	-	-
PC33. check oil temperature indicator (OTI) and winding temperature indicator (WTI)	1	1	-	-
PC34. check air-release plugs of main tank, radiator, conservator, bushings, etc. are free of air pocket / bubbles	-	2	-	-
PC35. energize distribution transformer at NO-LOAD only and checked for any abnormalities for the next 4 to 8 hours	2	2	-	-
PC36. take advice from the manufacturer or suppliers if any major abnormalities or defects found during repair and maintenance	2	3	-	-
<i>Testing and checking of over hauled Distribution transformer before delivery</i>	7	12	-	-
PC37. ensure complete transformer with its components are fitted and packed in its original shape	1	2	-	-
PC38. confirm all the test are done before delivery. all the test relevant to the performance of DT and ensure basic parameters like physical: leakage, low oil, silica in breather, HV & LV bushing. electrical: IR Value (HT to e, IT to e, HT to IT, Oil BDV)	2	2	-	-
PC39. ensure vent pipe is sealed with aluminum foil (diaphragm), temperature gauge is fitted and all HV terminals are fitted with horn and double screws and washers	-	2	-	-



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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC40. check list before delivery: oil level, no leakage of oil, tap position, silica gel in breather, radiator valve, thermometer packet, earth connection	1	2	-	-
PC41. ensure that the inspected and tested component meets the specified operating conditions before issue of ok certificate	2	2	-	-
PC42. anticipate problems well in advance in order to rectify timely	1	2	-	-
NOS Total	26	74	-	-



Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	PSS/N3006
NOS Name	Repair, Overhaul and Delivery of tested distribution transformer
Sector	Power
Sub-Sector	Distribution
Occupation	Distribution Transformer Repair & Testing
NSQF Level	4
Credits	TBD
Version	1.0
Last Reviewed Date	19/07/2016
Next Review Date	31/03/2022
NSQF Clearance Date	09/04/2018

Assessment Guidelines and Assessment Weightage

Assessment Guidelines

1. Criteria for assessment for each Qualification Pack will be created by the Sector Skill Council. Each Element/ Performance Criteria (PC) will be assigned marks proportional to its importance in NOS. SSC will also lay down proportion of marks for Theory and Skills Practical for each Element/ PC.
2. The assessment for the theory part will be based on knowledge bank of questions created by the SSC.
3. Assessment will be conducted for all compulsory NOS, and where applicable, on the selected elective/option NOS/set of NOS.
4. Individual assessment agencies will create unique question papers for theory part for each candidate at each examination/training center (as per assessment criteria below).
5. Individual assessment agencies will create unique evaluations for skill practical for every student at each examination/ training center based on these criteria.
6. To pass the Qualification Pack assessment, every trainee should score the Recommended Pass % aggregate for the QP.
7. In case of unsuccessful completion, the trainee may seek reassessment on the Qualification Pack.



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Minimum Aggregate Passing % at QP Level : 70

(Please note: Every Trainee should score a minimum aggregate passing percentage as specified above, to successfully clear the Qualification Pack assessment.)

Assessment Weightage

Compulsory NOS

National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
PSS/N1336.Working effectively with others	30	70	-	-	100	5
PSS/N2001.Use basic health and safety practices for power related work	24	76	-	-	100	15
PSS/N3005.Testing and inspection of various faults in distribution transformer	22	78	-	-	100	40
PSS/N3006.Repair, overhaul and delivery of tested distribution transformer	26	74	-	-	100	40
Total	102	298	-	-	400	100

Qualification Pack

Acronyms

NOS	National Occupational Standard(s)
NSQF	National Skills Qualifications Framework
QP	Qualifications Pack
TVET	Technical and Vocational Education and Training
LA	Lightening Arrestor
LT	Low Tension
LV	Low Voltage
MCB	Miniature Circuit Breaker
N	Neutral
OCB	Oil Circuit Breaker
OLTC	On Load Tap Changer
O/C	Over Current
O/H	Over Head
O&M	Operation & Maintenance
P	Phase / Power
PF	Power Factor
PT	Potential Transformer
PV	Photo Voltaic
PVC	Poly Vinyl Chloride
REC	Rural Electrification Corporation
SEB	State Electricity Board
SERC	State Electricity Regulatory Commission
T&D	Transmission and Distribution
T/F	Transformer
V	Voltage



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VT	Voltage Transformer
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Qualification Pack

Glossary

Sector	Sector is a conglomeration of different business operations having similar business and interests. It may also be defined as a distinct subset of the economy whose components share similar characteristics and interests.
Sub-sector	Sub-sector is derived from a further breakdown based on the characteristics and interests of its components.
Occupation	Occupation is a set of job roles, which perform similar/ related set of functions in an industry.
Job role	Job role defines a unique set of functions that together form a unique employment opportunity in an organisation.
Occupational Standards (OS)	OS specify the standards of performance an individual must achieve when carrying out a function in the workplace, together with the Knowledge and Understanding (KU) they need to meet that standard consistently. Occupational Standards are applicable both in the Indian and global contexts.
Performance Criteria (PC)	Performance Criteria (PC) are statements that together specify the standard of performance required when carrying out a task.
National Occupational Standards (NOS)	NOS are occupational standards which apply uniquely in the Indian context.
Qualifications Pack (QP)	QP comprises the set of OS, together with the educational, training and other criteria required to perform a job role. A QP is assigned a unique qualifications pack code.
Unit Code	Unit code is a unique identifier for an Occupational Standard, which is denoted by an 'N'
Unit Title	Unit title gives a clear overall statement about what the incumbent should be able to do.
Description	Description gives a short summary of the unit content. This would be helpful to anyone searching on a database to verify that this is the appropriate OS they are looking for.
Scope	Scope is a set of statements specifying the range of variables that an individual may have to deal with in carrying out the function which have a critical impact on quality of performance required.



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Knowledge and Understanding (KU)	Knowledge and Understanding (KU) are statements which together specify the technical, generic, professional and organisational specific knowledge that an individual needs in order to perform to the required standard.
Organisational Context	Organisational context includes the way the organisation is structured and how it operates, including the extent of operative knowledge managers have of their relevant areas of responsibility.
Technical Knowledge	Technical knowledge is the specific knowledge needed to accomplish specific designated responsibilities.
Core Skills/ Generic Skills (GS)	Core skills or Generic Skills (GS) are a group of skills that are the key to learning and working in today's world. These skills are typically needed in any work environment in today's world. These skills are typically needed in any work environment. In the context of the OS, these include communication related skills that are applicable to most job roles.
Electives	Electives are NOS/set of NOS that are identified by the sector as contributive to specialization in a job role. There may be multiple electives within a QP for each specialized job role. Trainees must select at least one elective for the successful completion of a QP with Electives.
Options	Options are NOS/set of NOS that are identified by the sector as additional skills. There may be multiple options within a QP. It is not mandatory to select any of the options to complete a QP with Options.